

Final Project: Classifying Online Job Postings into Occupations with LLMs

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Leander Müller (01/1589827), Philipp Raumanns (01/1590830),

Lars Stützle (01/1163443), Leon Exeler (01/1588945)

The goal of the present project was to implement automated online job-ad classification according to level 3 of the International Standard Classification of Occupations from 2008 (ISCO-08) using large language models (LLMs). To this end, we followed two approaches: The first approach comprised an in-context learning model, which we gave the job-ad and relevant level-3 ISCO definitions in the prompt without any training. The second approach was a two-step fine-tuned model, which we (1) pretrained the model on the level-3 ISCO-08 definitions based on next-token prediction and (2) instruction-fine-tuned on a small set of hand-labelled job ads. Comparing these two strategies, we wanted to find out down to which hierarchy level within the ISCO-08 taxonomy LLMs can reach reliable results. Subsequently, we used the best-performing model to label a large corpus of online job-ads from the US and conducted a labor market analysis based on this data.

Data

ISCO-08

The International Standard Classification of Occupations (ISCO) is an occupation classification framework aiming to make occupations internationally comparable. We used ISCO-08, which is its current 2008 version. It is published by the International Labor Organization (ILO) and available under the following link: <https://ilostat.ilo.org/methods/concepts-and-definitions/classification-occupation/>

The ISCO-08 taxonomy hierarchically groups occupations on four levels. It includes 10 major groups at level 1, which are further subdivided into 43 sub-major groups at level 2, 130 minor groups at level 3, and 436 unit groups at level 4. Across levels, each group includes a title, a detailed definition and a code, with the number of digits corresponding to the level. For example, the unit group Software Developers (2512) is part of the minor group Software and Applications Developers and Analysts (251) on level 3, the sub-major group Information and Communications Technology Professionals (25) on level 2, and the major group Professionals (2) on level 1.

In the present project, we used the ISCO-08 minor groups, corresponding to level 3 of the ISCO-08 taxonomy. The information we used for each minor group, both for the prompts for the in-context learning model and the continued pretraining of the two-step fine-tune models, was structurally identical. Specifically, we constructed a string including the group titles at level 1, level 2, and level 3, as well as the definition of the level 3 minor group.

Job-ad Corpus

As job-ad corpus, we used an open dataset of initially 33,246 Linked-In job ads posted in the US. The dataset is publicly available via the Hugging Face Hub under the following link: <https://huggingface.co/datasets/xanderios/linkedin-job-postings>. Out of the available information, we decided to use the title, the description, the job location and the date the ad was published. Job locations varied in their level of detail, referring to cities, metropolitan areas, states, or countries (i.e., the US). Thus, we categorized them on state-level and labeled all locations indicated on country-level as unknown. After removing duplicates and job-ads with no description, a total of 30,587 observations remained, published between August 10 and November 4, 2023.

To produce a cleaned and anonymized corpus suitable for subsequent analysis, we applied a preprocessing pipeline to remove personally identifiable information. First, we detected email addresses, phone numbers, and URLs using pattern-matching techniques and replaced them with placeholder tokens ([EMAIL], [PHONE], and [URL], respectively). Second, we applied named entity recognition (NER) using the spaCy natural language processing library (Honnibal et al., 2020) to identify and redact organization names and personal names, replacing them with [ORG] and [PERSON] tags accordingly.

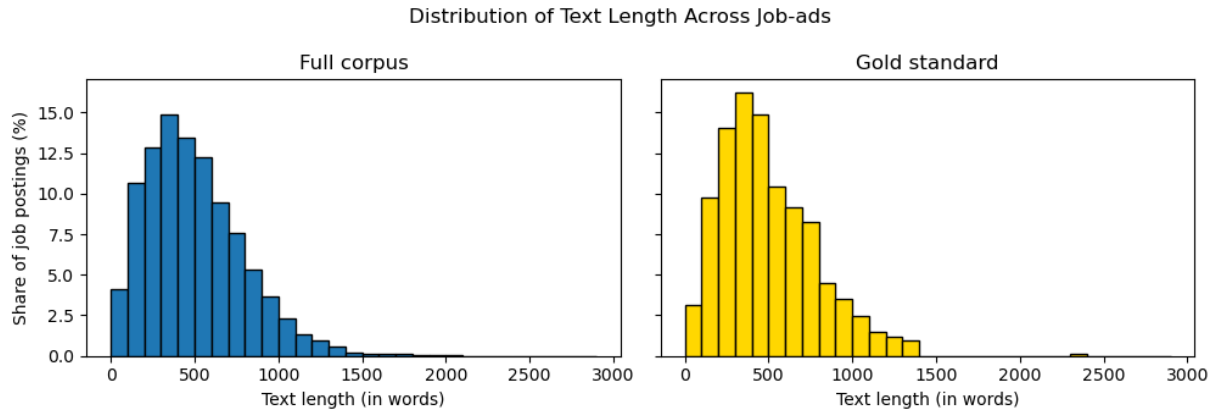
Gold Standard Set

From this job-ad corpus, we randomly retrieved a gold standard set of 740 job-ads, which we manually assigned ISCO-08 minor group codes, i.e., codes at level 3 of the ISCO-08 taxonomy. We split this data into four batches of 185 ads. Each of four group members labelled three batches, i.e., 555 ads in total. We distributed the batches such that each ad was labelled by three different annotators. We obtained a final ISCO-08 code label based on two cases: If two or more annotators opted for the same label, i.e., if there was a majority, then this label became the final label. If all annotators opted for a different label, we used an LLM to assign an ISCO-08 minor group label.

Preliminary Data Analysis

As illustrated in Figure 1, the shape of the text length distribution was very similar across the full corpus and the gold standard set.

Figure 1



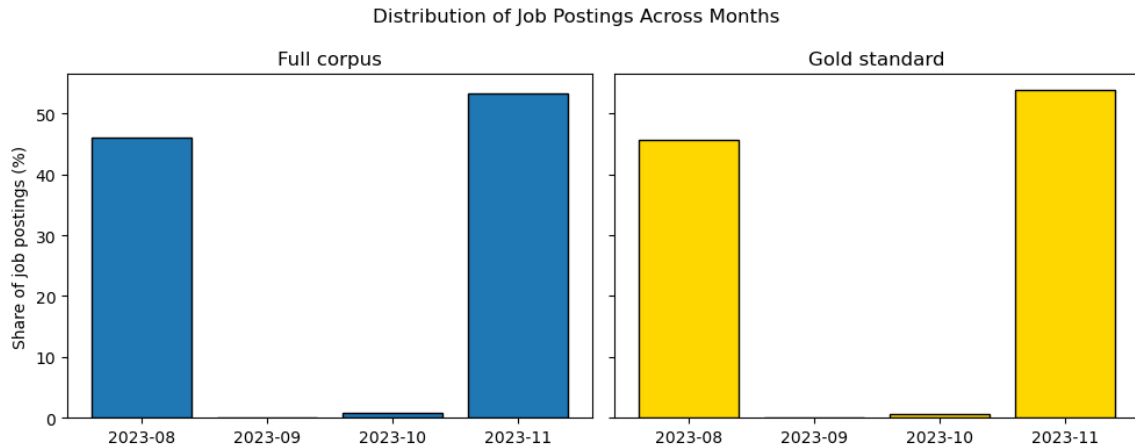
In both datasets, text lengths were on average at about 500 words and showed considerable variation (see Table 1). The main difference was that the full corpus had a larger maximum.

Table 1: Summary Statistics for Text Lengths in Words

Class	<i>M</i>	<i>SD</i>	Min	Max
Full corpus	499.53	294.65	1	2839
Gold standard	497.71	286.39	5	2307

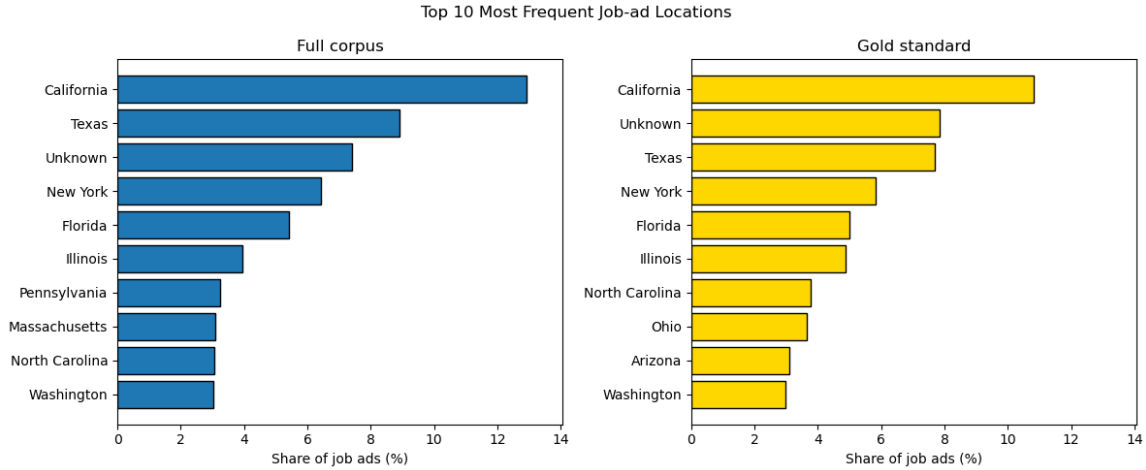
Also in terms of the temporal (see Figure 2) and spatial (see Figure 3) distributions of job-ads, both datasets were overall very similar. This suggests that the gold standard set is a representative sample of the full corpus. With regard to the publishing date, the job-ad data was heavily imbalanced. Job-ads were mainly posted either in August or November 2023, with very few observations in October and none in September.

Figure 2



With regard to job location, there was also considerable imbalance in the data. Five states accounted for substantially larger shares of online job postings than the others, with California leading by a wide margin. The geographic distribution of job postings was concentrated in economically large states, particularly California and Texas, which likely reflects both the size of their labor markets and their geographic coverage.

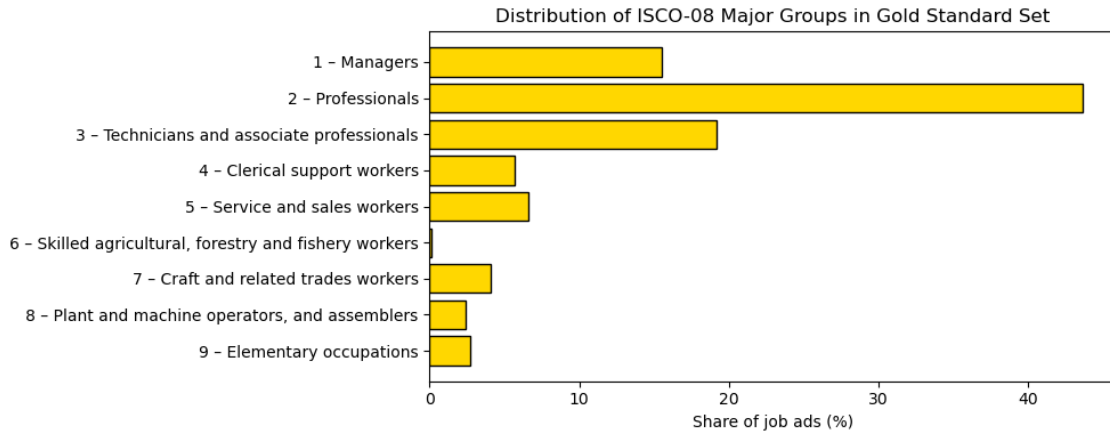
Figure 3



Note. The category “unknown” represents US locations for which the state was not indicated.

The distribution of ISCO-08 major groups in the gold standard set (see Figure 4) shows a clear dominance of professionals in our data, accounting for more than 40% of job-ads. Trailing well behind, the major groups Technicians and Associate Professionals (19%) and Managers (15%) follow. All remaining groups occurred considerably less frequent, each accounting for approximately 5% of job ads or less. The group of skilled agricultural, forestry and fishery workers only occurred once, while the group of armed force occupation was not represented at all. Overall, the ISCO-08 codes were heavily imbalanced in the gold standard set, indicating that the full job corpus may exhibit a similarly uneven occupational distribution.

Figure 4



Next, we assessed the quality of the ISCO-labels in the gold standard set. For each ISCO-08 level until level 3, we measured inter-annotator agreement as the percentage of job-ads in the gold standard set labelled with the same ISCO-08 category by all annotators (3/3), labelled with the same ISCO-08 category by the majority of annotators (2/3), or labelled with a different ISCO-08 category by all annotators (1/3). Additionally, we computed Fleiss’ κ , which provides a valuable extension by accounting for the agreement expected by chance.

Table 2 shows the results of this analysis. Inter-annotator agreement decreased with increasing ISCO-08 level, i.e., as the classification becomes more fine-grained. Level 1 showed good results, including only about 2% of job-ads for which no majority label was present among the annotators themselves. For level 3, the share of these job-ads was considerably higher. Together with the fact that for only around half of the job ads, all annotators agreed on a label, this suggests that the level-3 labels should be treated with some caution. This pattern is also reflected in the Fleiss’ κ values, which decreased from 0.75 at Level 1 to 0.67 at Level 3. These values indicate substantial inter-annotator agreement across all three levels, although agreement becomes weaker at the more granular level 3.

Table 2: Inter-annotator Agreement per ISCO-08 Level

ISCO-08 level	3/3	2/3	1/3	Fleiss’ κ
Level 1 (major groups)	73.38%	24.46%	2.16%	0.75
Level 2 (sub-major groups)	64.86%	30.95%	4.19%	0.73
Level 3 (minor groups)	55.95%	36.22%	7.84%	0.67

To examine the aspects that might have contributed to these apparent job-ad labelling difficulties, we inspected the in total 58 job-ads, for which all annotators provided different ISCO-08 categories. First, vague or broad job titles such as “Lead Analyst” or “Assistant Manager” provided insufficient information about the actual occupational content. Second, multi-role job-ads combined tasks from different occupational domains, making it difficult to identify the primary occupation. Third, marketing-oriented and company-specific language often emphasized soft skills, culture, and broad responsibilities rather than concrete occupational tasks.

In-Context Learning Model

Classifying the job-ads into ISCO-08 codes via an in-context LLM was structured into three steps. First, we generated semantic vector representations using the BAAI/bge-en-icl embedding model (Li et al., 2025). We chose an asymmetric retrieval approach. We embedded ISCO category profiles (containing titles, definitions, and associated tasks) as raw passage documents, while we prepended job postings with the task instruction (“Represent this sentence for searching relevant passages: ”). We L_2 -normalized all embedding vectors to compute pairwise cosine similarities via dot product. The top-k candidate profiles were retrieved and randomly shuffled to eliminate position bias. Second, we compiled the job descriptions, their candidate options, and a system prompt establishing an expert labour market analyst persona into a standardized chat template. Instructions constrained the model to select the single best-matching candidate, providing a one-sentence rationale, and conclude with a fixed closing pattern: “Therefore, I chose Candidate: [Index]”. Third, we executed inference using meta-llama/Llama-3.1-8B-Instruct in bfloat16 precision (Grattafiori et al., 2024). We prompted the model the standardized chat templates. To ensure deterministic and reproducible outputs, we applied greedy decoding (do_sample = False) with a generation limit of max_new_tokens = 200. From the generated output, we extracted the final decision via regular expressions.

During model tuning, initial runs with BAAI/bge-large-en-v1.5 (Xiao et al., 2023) revealed that roughly 68% of job-ads exceeded its 512-token limit (mean: 681.1 tokens; maximum: 1,720 tokens). This meant information loss due to truncation and hence performance loss in retrieving the gold label ISCO code. To counter this, we switched to the next larger BGE model: BAAI/bge-en-icl (Li et al., 2025), which natively supports context windows up to 32k tokens. Moreover, in early iterations where the decoder LLM was tasked with outputting raw 4-digit ISCO codes directly, the model frequently hallucinated non-existent codes or codes outside the given candidate pool. Restricting the decision space to single-token candidate indices (1 to k) enforced a strict closed-world classification constraint, raising the instruction compliance rate to 100.0%. To distinguish between calculated conditional accuracy measures, retrieval bottlenecks (i.e., the correct class not being retrieved) and reasoning errors (i.e., the LLM selecting the wrong option among available candidates), we constructed an additional oracle candidate set. For each job posting where the ground-truth label was missing in the top-k retrievals, the true ISCO class was inserted into the candidate pool. Both sets were then prompted to the decoder LLM. Finally, we conducted parameter sweep for the size of the candidate pool (k = 1 up to 20) to evaluate the trade-off between embedding recall and prompt complexity. K = {5, 10, 15} were then selected for full decoder evaluation to assess whether increasing candidate-pool size improved retrieval at the expense of downstream classification performance.

Table 3 shows the result of the evaluation carried out across all ISCO-08 hierarchy levels on the validation set (N=104). We selected k = 5 as the best configuration based on the highest level-3 macro F1 and accuracy across all hierarchy levels. Although larger candidate pools increased retrieval recall, they did not improve downstream classification performance, suggesting that additional candidates introduced distractor noise.

Table 3: Comparison of Model Performance on Validation Set for Different k

k	Retrieval Recall	F1 Level 3	Acc. level 3	Acc. level 2	Acc. level 1
5	78.85%	0.380	73.08%	63.46%	55.77%
10	87.50%	0.377	68.27%	58.65%	50.00%
15	91.35%	0.327	59.62%	50.96%	46.15%

Note. Acc.: accuracy.

Two-Step Fine-Tuned Model

For the two-step fine-tuned LLM, we used the meta-llama/Llama-3.1-8B model (Grattafiori et al., 2024) and LoRA adapters from the peft library (Mangrulkar et al., 2022). Due to the limited amount of labelled data, a full fine-tuning approach was not feasible.

The training was split into two steps. Step 1, the continued pretraining, focused on internalizing the ISCO-08 taxonomy. The training corpus consisted of the official ISCO-08 minor-group descriptions. We expanded the 130 categories into 260 next-token-prediction examples by creating both a forward mapping (code to description) and a backward mapping (description to code). We split this data into 234 training and 26 evaluation samples (90/10 split). The, we applied LoRA to all attention and MLP projections with

rank $r = 8$, $\alpha = 32$, and a dropout of 0.1. We trained the model for 8 epochs at a learning rate of 1×10^{-4} with a cosine schedule, using an effective batch size of 16 and early stopping (patience 3) on the validation loss. During training, both training and validation loss showed a steep and continuous decrease, with validation loss flattening out in the last epochs (see Figure 5, left plot). We retained the checkpoint with the lowest validation loss.

For Step 2, the instruction fine-tuning, we merged the step 1 adapter into the base model. Then, we trained a second LoRA adapter on the labelled gold standard job-ad set. This adapter targeted the same modules with the same rank ($r = 8$) but used $\alpha = 16$. We formatted the inputs as instructions mapping the job ad title and text to the corresponding 3-digit ISCO code, with a maximum sequence length of 2048 tokens. This step was also trained for 8 epochs at a learning rate of 1×10^{-4} with an effective batch size of 16 and early stopping. The resulting checkpoint is referred to as run_11.

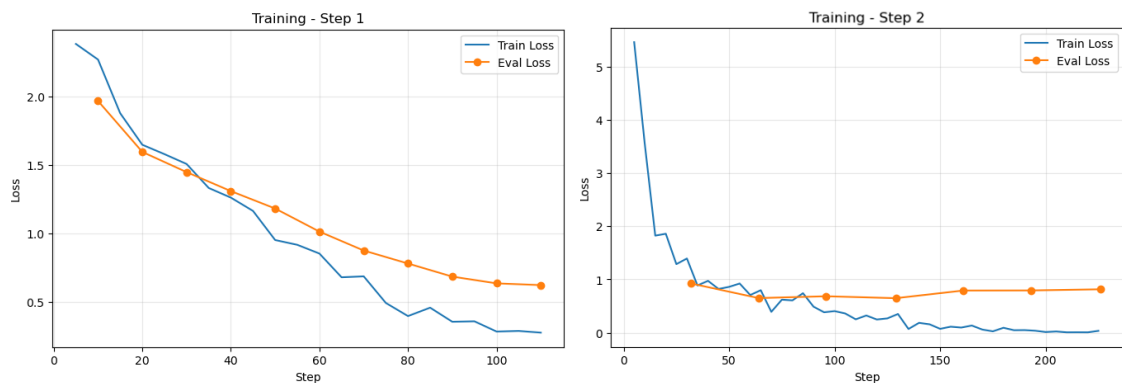
To find the best configuration for Step 2, we performed a sequential (greedy) hyperparameter search instead of a full grid search. By varying one parameter at a time and keeping the best-performing setting as the baseline for the next step, we could explore the parameter space much more efficiently. Model selection at each phase prioritized the evaluation-set macro F1 score, which was more informative than raw accuracy given the strong class imbalance across the 85 represented ISCO codes. Accuracy served as a secondary metric. Table 4 summarizes the search process.

Table 4: Results of Sequential Hyperparameter Search on the Evaluation Split

Phase	Run	Modification	3-dig. Acc. (%)	Macro F1	Outcome
0: Baseline	run_01	$r = 32$, $\alpha = 64$, LR = $1e-4$	69.2	0.632	Baseline
1: Learning rate	run_02	LR = $5e-5$	67.3	0.557	Rejected
	run_03	LR = $5e-4$	69.2	0.548	Rejected
2: LoRA rank	run_04	$r = 8$, $\alpha = 16$	75.0	0.672	Accepted
	run_05	$r = 64$, $\alpha = 128$	76.0	0.634	Rejected
3: Step 1 ablation	run_06	Step 1 pretraining OFF	67.3	0.575	Rejected
4: Max. seq length	run_07	max_length = 1024	64.4	0.503	Rejected
5: Eff. batch size	run_08	eff. batch = 8	75.0	0.637	Rejected
	run_09	eff. batch = 32	71.2	0.589	Rejected
6: Epochs	run_10	epochs = 10	61.5	0.467	Rejected

The only modification that improved the baseline was adjusting the LoRA rank and alpha in Phase 2 (run_04). Conversely, removing the step 1 pretraining (run_06) led to a substantial performance drop, reducing 3-digit accuracy from 75.0% to 67.3% and macro F1 from 0.672 to 0.575. This confirms that the internalization of the taxonomy during step 1 provides a meaningful benefit to the model’s classification ability at this configuration. Therefore, we retrained the final configuration (run_11) from scratch using all accepted settings. During the first training epochs, both training and validation loss decreased rapidly, followed by a substantially slower decline in training loss and a slight increase in validation loss, suggesting some overfitting (see Figure 5, right plot). We therefore retained the checkpoint with the lowest validation loss. Before deployment, we re-ran the evaluation split using the original random seed. The loaded run_11 adapter reproduced the training-time accuracy of 74.0% and macro F1 of 0.673, confirming that the saved model functioned correctly.

Figure 5



Model Comparison and Evaluation

Table 5 compares the performance of the final in-context learning and two-step fine-tuned models on the validation set. While for both models, accuracy increased as the ISCO-08 level became coarser, the

two-step fine-tuned model clearly outperformed the in-context learning model across all metrics. Most remarkably, the macro F1 score for level 3 was almost twice as high. While the two-step fine-tuned model clearly yielded better performance, this came at the cost of additional computational resources and training time, as well as the need to manually create a labeled gold standard for fine-tuning. In contrast, the in-context learning approach required no model training and can therefore be applied more easily to new classification tasks. Nevertheless, given the substantial performance advantage of the two-step fine-tuned model, particularly at the most fine-grained level, we selected it as our final model.

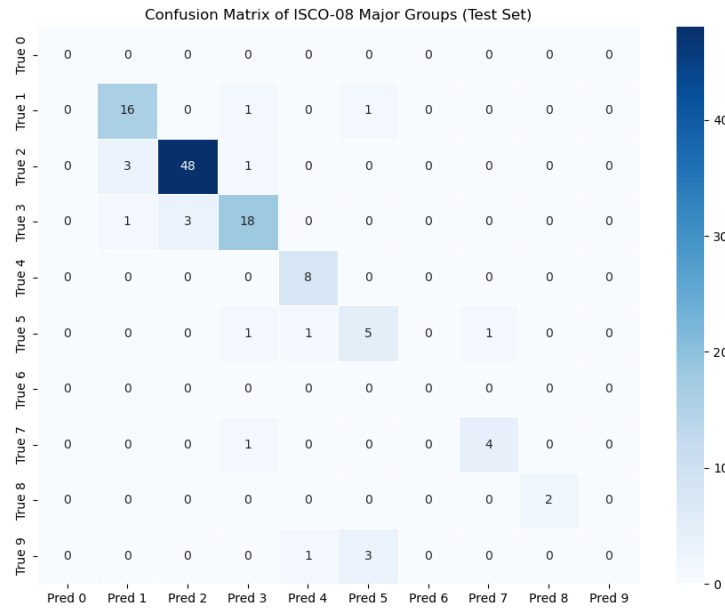
Table 5: Model Performance on Validation and Test Set

Split	Model	F1 Level 3	Acc. level 3	Acc. level 2	Acc. level 1
Validation	ICL	0.38	49.04%	53.85%	61.54%
Validation	2SFT	0.67	74.0%	81.7%	84.6%
Test	2SFT	0.58	73.1%	79.0%	84.9%

Note. ICL: in-context learning model, 2SFT: two-step fine-tuning model, Acc.: accuracy.

The performance of the two-step fine-tuned model on the test set (also see Table 5), was comparable. While the accuracies remained stable, the macro F1 score dropped to 0.58. Because macro F1 weights every class equally, this drop was largely driven by a few rare classes that had zero correct predictions in the test set. Given the small sample size of 740 ads spread across 85 codes, this likely reflects a lack of data for rare classes rather than the test set being inherently more difficult. Model accuracy improved as the classification granularity was relaxed. This confirms the intended hierarchy-aware behavior. Most fine-grained errors were near-misses that landed in the correct or a related major group, rather than random misclassifications. This is corroborated by the confusion matrix (Figure 6), showing that off-diagonal misclassifications occur almost entirely between neighboring major groups. For example, Professionals (group 2) are mainly confused with adjacent groups like Managers (group 1) and Technicians (group 3).

Figure 6



Model deployment

The final model was deployed to the job-ads in the full corpus, that were not part of the already labeled gold standard set. It included $30,587 - 740 = 29,847$ job-ads, for which the model predicted the ISCO-08 level 3 code (minor group) based on the job-ad text. Only two predictions failed to produce a valid ISCO code, confirming the 100% valid-code rate observed during testing. To sanity check the model's predictions, we conducted two complementary analyses: a manual spot-check of individual predictions and a comparison of the predicted occupational distribution with external occupational statistics.

We manually spot-checked 10 random predictions from the full corpus. All generated codes were valid, and straightforward roles were mapped as expected, e.g., Physical Therapist to 226 and AI Architect to 251. More complex postings showed that the model reacts to the full job ad text and not just the title: a Public Safety Technical Functional Analyst was mapped to Database and Network Professionals (252) rather than Software and Applications Developers and Analysts (251), since the text focused mainly on server maintenance and firewall administration; a Brand Ambassador Lead was mapped to Business Services Agents (333) rather than a marketing code, since the text described event-security planning and staffing

rather than promotional work. A closer look at these and similar cases suggests the model may underweight seniority terms such as Director or Lead in favor of the subject-matter domain. A Sr Director Biostatistics posting, for example, emphasized staff leadership, budgets, and departmental strategy, yet was classified into a life-science code rather than a managerial one. We have no ground truth for the unlabeled corpus, so we cannot confirm whether these predictions are right or wrong, but this points to a specific and plausible failure mode rather than proof that the model always reasons correctly.

As a second sanity check, we compared the predicted major group distribution against the US Bureau of Labor Statistics (BLS) May 2023 OEWS data (U.S. Bureau of Labor Statistics, 2024), mapped via the official ISCO-08/2010 SOC crosswalk (U.S. Bureau of Labor Statistics, 2012). The results are shown in Table 6. Where a single SOC code maps to multiple ISCO-08 codes, employment is split equally across the linked codes, per the BLS crosswalk documentation. Thus, the predicted-vs.-BLS gaps below should be read with this simplification in mind.

Table 6: Predicted Major Group Distribution vs. US BLS Employment Shares

Major Group	Predicted Share (%)	US BLS Share, 2023 (%)	Diff
0	0.0	0.03	−0.03pp
1	15.7	7.3	8.3pp
2	44.1	19.1	25.0pp
3	19.4	16.8	2.6pp
4	5.8	11.7	−5.9pp
5	7.3	18.1	−10.8pp
6	0.3	0.3	0.1pp
7	3.9	8.9	−5.1pp
8	1.9	6.1	−4.2pp
9	1.6	11.7	−10.0pp

Group 0 (Armed Forces) receives zero predictions across all 29,847 ads, consistent with it having no representation in the test split; its reliability is entirely untested rather than confirmed to be poor.

Our dataset heavily over-represented Professionals (group 2) and Managers (group 1) while underrepresenting elementary (group 9), service (group 5), and clerical occupations (group 4). Two factors explain this divergence. First, the corpus is sourced from LinkedIn job postings, a platform that inherently skews toward white-collar and professional roles, so some of this gap reflects the true composition of the data source rather than the broader US economy. Second, as shown by the 0% recall for group 9, classifier bias systematically misroutes elementary-occupation ads into other groups. Thus, the predicted shares for groups 5 and 9 should be viewed strictly as lower bounds rather than exact estimates, compounding rather than replacing the platform-composition effect.

As reliability analysis, we examined model performance across occupations in the labeled test set. Performance varied considerably across ISCO major groups. As Table 7 shows, reliability is strongly related to how well a group is represented in the data, both in the test split shown here and, by extension, in training.

Table 7: Per-Major-Group Precision, Recall and F1 on the Test Set

Major Group	Precision	Recall	F_1	Support (test)
0	—	—	—	0
1	0.800	0.889	0.842	18
2	0.941	0.923	0.932	52
3	0.818	0.818	0.818	22
4	0.800	1.000	0.889	8
5	0.556	0.625	0.588	8
6	—	—	—	0
7	0.800	0.800	0.800	5
8	1.000	1.000	1.000	2
9	0.000	0.000	0.000	4

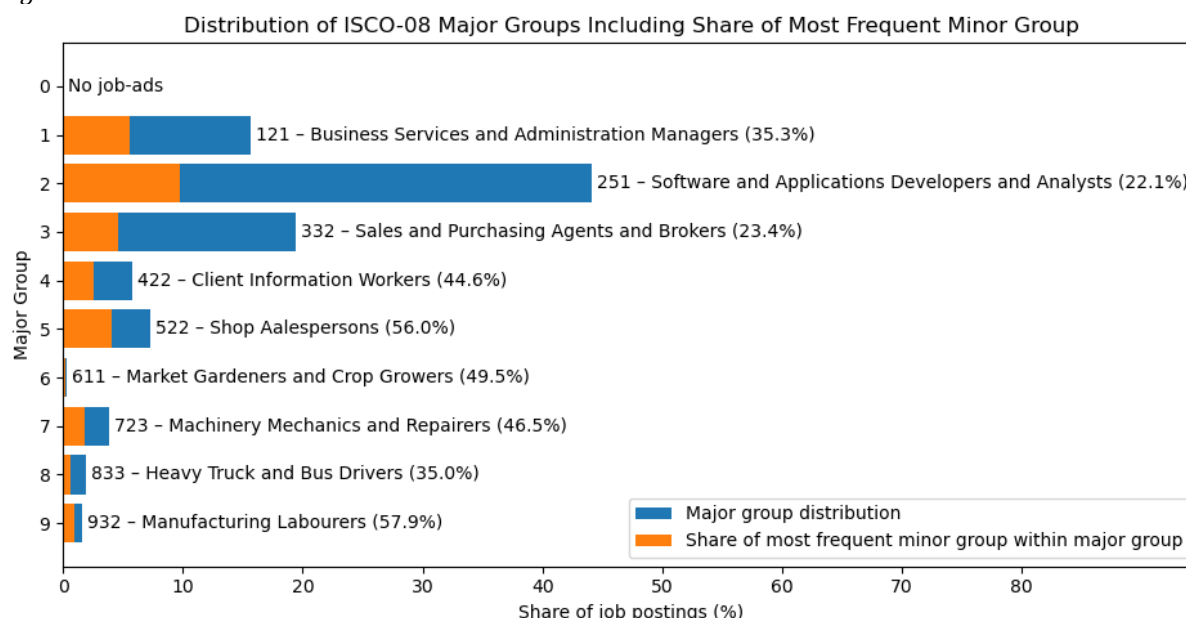
The best-represented groups (1, 2, and 3) were classified most accurately. Groups with very few training examples, such as 5 and 9, performed poorly. This indicates a data-scarcity problem: pretraining on the ISCO manual (step 1) is not a full substitute for actual labelled examples during fine-tuning. Hence, when interpreting the final labelled corpus, labels assigned to the reliable groups (1, 2, and 3) can be used with higher confidence, while labels in groups 5, 7, and 9 should be treated with caution.

Labor Market Analysis

We used the labelled corpus of 29,847 US online job-ads to gain insights on the composition of job demand in the US labor market. The distribution of ISCO-08 major groups, as illustrated in Figure 7, closely reproduced the pattern observed in the gold standard set. Professionals remained by far the most prevalent

group, followed by technicians and associate professionals and managers, while all other major groups accounted for substantially smaller shares. As in the gold standard set, skilled agricultural, forestry and fishery workers were only marginally represented, and armed forces occupations were absent entirely. With regard to the most frequent minor groups within each major group, the less frequent major groups were often dominated by a single occupation, accounting for around 50% of job-ads. This share was lower for the three most frequent major groups, suggesting a more diverse occupational composition within these groups.

Figure 7



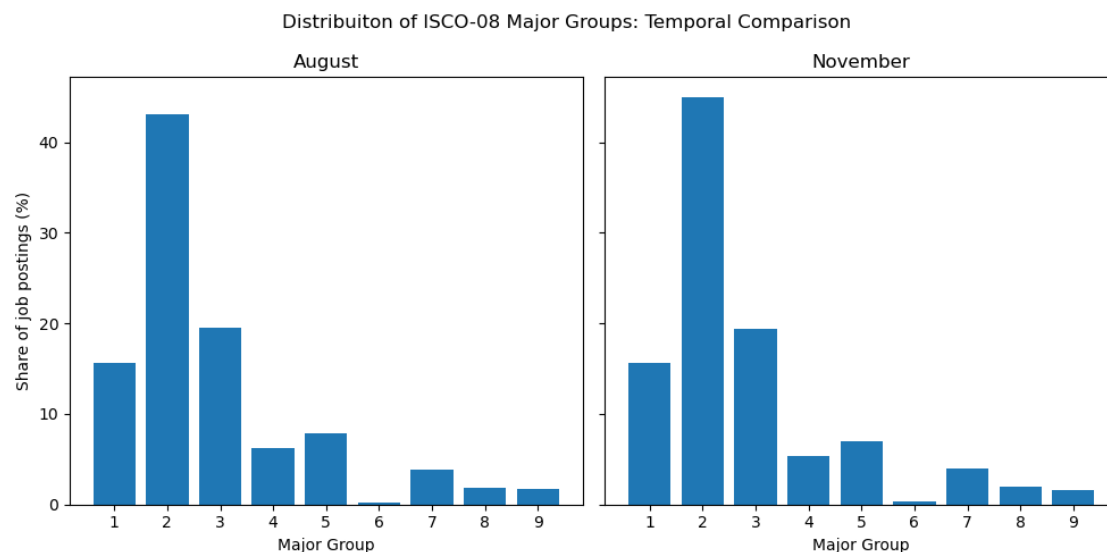
The ten most frequent minor groups (see Table 8) aligned with the major group distribution, being heavily dominated by white-collar professional and managerial roles, such as software developers, nursing professionals, and finance experts.

Table 8: Top 10 Most Frequent ISCO-08 Minor Groups

ISCO Code	Title	Count
251	Software and applications developers and analysts	2,904
121	Business services and administration managers	1,649
222	Nursing and midwifery professionals	1,483
214	Engineering professionals (excluding electrotechnology)	1,458
241	Finance professionals	1,360
332	Sales and purchasing agents and brokers	1,358
243	Sales, marketing and public relations professionals	1,297
522	Shop salespersons	1,224
132	Manufacturing, mining, construction and distribution managers	1,064
242	Administration professionals	1,051

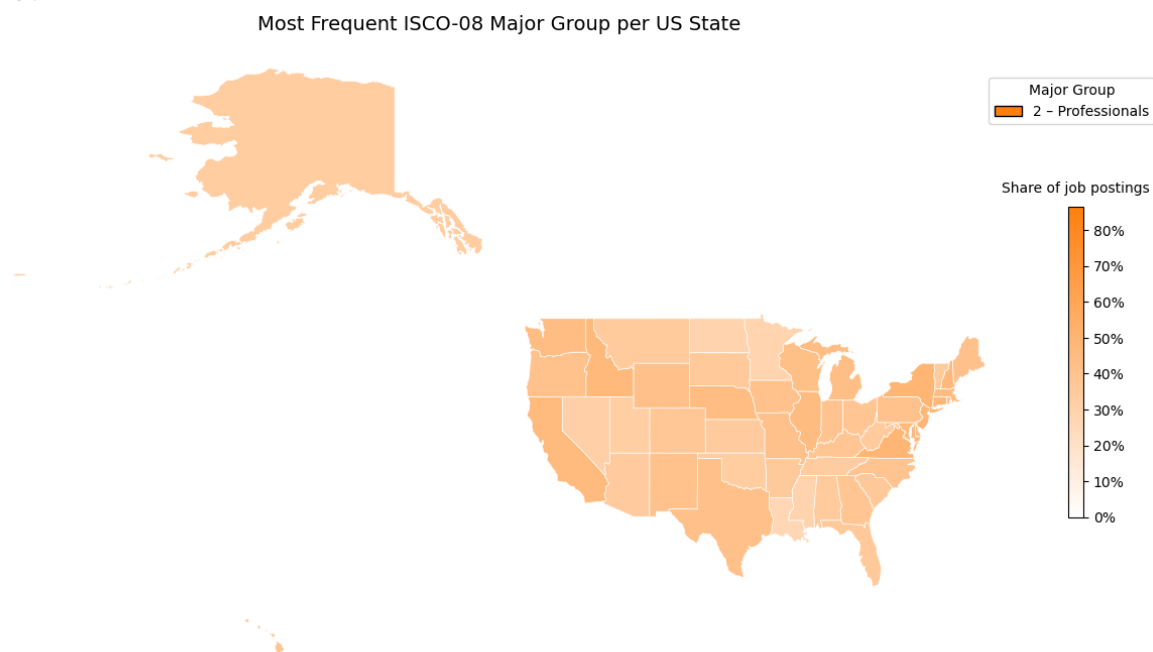
The major group distribution appeared temporally stable. As Figure 8 indicates, there were virtually no differences in major group composition of US online job-ads between August 2023 and November 2023, i.e., the first and last months of our observation period.

Figure 8



When comparing job-ad composition geographically, major group 2, the Professionals, was the most frequent major group in all states (see Figure 9). However, its degree of its dominance varied to some extent across states: While its share was particularly high in economically strong regions like California, the Northeast, and Texas, reaching almost 50% in some states, it was generally lower in more rural states, where it typically accounted for around 30-40% of job ads. This suggests that although professional occupations dominate online job demand across the US, the occupational composition of online postings still varies geographically, potentially reflecting differences in regional economic structure.

Figure 9



While our analysis provides insights into occupational demand as represented in online job-ad data, several limitations constrain its interpretation as a representation of the broader US labor market. First, coverage bias is likely to overstate the prevalence of major groups 1-3 in the overall market. This is because these groups may be disproportionately advertised online, while occupations in other groups may be more commonly recruited through offline channels or alternative hiring mechanisms. Second, our corpus is drawn from a single online platform, LinkedIn, and is therefore not necessarily representative of US online job-ads as a whole. Specifically, LinkedIn may attract a particularly high share of professional, managerial, and technical occupations, while other occupational groups may be better represented on other job platforms. Consequently, the observed distribution should be interpreted as a distribution of LinkedIn online job postings rather than a direct measure of actual US job demand. Finally, regional differences in platform usage

and recruitment practices may contribute to the geographic variation observed in Figure 9. Differences between states may therefore not only reflect differences in occupational demand, but also differences in which employers use online platforms and how they recruit workers.

Conclusions

Overall, the results show that LLMs can successfully support automated occupational classification of online job ads, but that performance decreases with increasing classification granularity. The two-step fine-tuned model clearly outperformed the in-context learning approach and benefited substantially from continued pretraining on the ISCO-08 taxonomy. At the same time, performance varied across occupations, with rare major groups being classified less reliably due to limited labelled training data. Applied to the full corpus, the model produced plausible classifications and reproduced the occupational distribution observed in the gold standard set, with Professionals, Technicians, and Managers accounting for the largest shares. The resulting labor market analysis further indicated a stable occupational composition over the observation period and geographic variation in the prevalence of professionals, the most frequent major group. However, these findings were limited online job-posting frequencies on a single platform rather than representing the US labor market as a whole and should be interpreted with caution.

Automated occupational coding in official statistics and research offers substantial practical benefits, particularly in terms of scalability and efficiency, as large numbers of job ads can be classified much faster than through manual coding. However, differences in reliability across occupations require validation against manually labelled data and, where appropriate, human review of rare or ambiguous cases. For use in official statistics and research, transparent documentation and versioning of models, prompts, and training data are also essential for reproducibility. Ethically, particular attention must be paid to potential biases, as systematic misclassification of poorly represented occupations could distort findings about specific groups. In addition, job ads may contain personal or organizational information, making careful anonymization and secure data handling necessary. Finally, classification errors become particularly consequential when automated occupational codes are used to inform policy decisions or resource allocation, as systematic errors could lead to misleading conclusions or reinforce existing inequalities.

The present analysis comes with several limitations. First, the gold standard set was strongly imbalanced across ISCO occupations, with some minor groups represented by only very few job ads. This limits the model's ability to learn and reliably classify rare occupations. Future work could therefore use a larger and more stratified gold standard set to improve coverage of underrepresented occupations. Second, assigning some job ads to fine-grained ISCO categories was inherently ambiguous, for example when postings combined multiple roles or when the available information was insufficient to distinguish between closely related occupations. Future work could investigate more systematic annotation guidelines or allow multiple labels and uncertainty estimates for such cases. Third, the analysis is based on a single LinkedIn dataset and therefore reflects the composition of this particular source rather than the US labor market as a whole. Future work could combine multiple job-ad platforms and longer observation periods to assess the generalizability and temporal stability of the results.

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